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Deep Ensemble Learning for Early Crops Leaf Disease Classification and Risk Reduction

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***Abstract*—Plant diseases can significantly impact agricultural productivity and jeopardize a sufficient global supply of food. Therefore, early plant disease diagnosis reduces risks and losses. However, identifying and classifying plant leaf diseases by hand has become challenging and time-intensive, and raises the chance of farmer error. On the contrary, deep learning (DL) algorithms have demonstrated impressive outcomes when it comes to the early detection of diseases in plants. This study introduces a novel approach for categorizing plant leaf diseases to alleviate these limitations. The proposed approach creates a state-of-the-art ensemble model using Xception, MobilenetV3, and EfficientNetB7. This ensemble model incorporates a three-layer pre-train model that utilizes a coordinate attention block, softmax layer, reshape layers, dense layer, flattened layer, and a 60% dropout rate. To enhance the model performance, we employed data augmentation techniques like horizontal flips along with histogram equalization. In addition, this research evaluated the model performance with and without transfer learning (TL) and augmentation. Impressively, our proposed ensemble model, incorporating TL and augmentation, outperformed existing models with 98.70% accuracy, 99.10% precision, 98.70% recall, and 98.80% F1 score in classifying images of both normal and diseased leaves. Experimental results showed that our identification models adequately prevented agricultural diseases and ensured food security. Additionally, modern explainable artificial intelligence (XAI) utilizing gradient-weighted class activation mapping (Grad-CAM) can precisely identify crop-affected regions.**

***Index Terms*—Leaf Disease, Augmentation, Deep Learning, Transfer learning, Grad-CAM**

I. Introduction

Globally, plants exert a substantial influence on many aspects of civilization, including the weather, agriculture, and economy. A sustainable ecosystem and reliable food supply depend on the biodiversity of both animal and plant forms. Worldwide food production must be increased by 60% to meet the demands of this growing population [1]. Various environmental conditions and weather patterns significantly influence the quality of agricultural products and subsequently affect the economy [2]. Therefore, the earliest phase in improving the quality of food is to prevent diseases in the plants that grow it. Early disease detection on plant leaves can guarantee satisfactory agricultural production. However, classifying leaf diseases is challenging because of the high degree of similarity between classes and the complexity of changes in their designs. Sometimes farmers identify plant diseases by hand, often by physically inspecting plant leaves through visual inspection or consulting an expert; however, this method is inefficient because it is tedious, error-prone, time-consuming, and costly [3]. In this research, we propose an automated technique that instantly identifies sick leaves, thereby enabling the early detection of plant illnesses that cause an increase in crop production. Also, automated localization and classification of crop diseases remain a challenging problem owing to large differences in leaf size, shape, color, and location [4]. The fact that the colors of leaves change when images are taken complicates the process of finding them. Recently, Artificial intelligence (AI) and computer vision (CV) have enabled the identification of diseases without relying solely on human examinations. This study discusses a variety of methods for identifying plant leaf diseases, ranging from traditional methods to the latest advancements in DL approaches. It extracts significant information from images using kernels, which are also called filters. These derived traits can be used to identify various plant diseases.

By integrating multiple DL techniques, we developed a methodology that is highly effective in detecting and classifying plant infections. For this, we utilized a bespoke ensemble model along with several pre-trained models such as Xception, MobileNetV3, and EfficientNetB7. Combining these three pretrained models resulted in the creation of proposed ensemble model in the PlantVillage dataset, a group of DL models that classifies crop diseases.

The study contributions listed below are:

i. We present an advanced DL based ensemble method for crop disease classification.
ii. This ensemble model was created by merging three DL models using a coordinate attention block, reshaping layers, a Softmax layer, a dense layer, and a flattened layer with a dropout rate of 60%.
iii. The advanced XAI method Grad-CAM using the proposed ensemble model successfully detected the affected areas of crops such as tomato, potato, and pepper.

A review of the intended study is presented in Section 2, while the methodology is explained in Section 3. Section 4 provides an extensive evaluation of the experimental findings and evaluation. Lastly, the paper is wrapped up in Section 5, which includes future work suggestions and a discussion of its limitations.

II. Literature Review

Plant pathologists inspect various plant organs such as roots, kernels, stems, and leaves, to classify plant diseases. With the help of Grad-CAM, Khalid et al. [5] investigated the use of Convolutional Neural Networks (CNNs) and MobileNet architectures for the detection of plant diseases. Contrasted with MobileNet's 96% accuracy, which was lower than CNN's 89% recall, 90% precision, and 89% F1 score, the CNN model had 89% accuracy, 96% precision, and 96% recall. The SE-SK-CapResNet model for plant leaf disease classification was proposed by Zhang et al. [6] and combines a ResNet with a channel attention mechanism and CapsNet. Merging ResNet and CapsNet features allowed the model to attain impressive accuracy levels of 98.58%, 95.08%, and 97.19% according to several evaluation metrics. Radočaj et al. [7] suggested IncMB, a module that combines the inception module, Mish activation, and batch normalization, to find diseases in tomatoes. They evaluated CNNs, CNNs+SVM, and CNNs+IncMB with PlantVillage, which has six disease classifications. The pre-trained InceptionV3 with IncMB had the best accuracy of 97.78%, which was better than most standard CNNs. Building on PlantVillage, Sadhasivam et al. [8] suggested a 10-layer LWR network for plant disease diagnosis. This network would have convolutional filters ranging from 32 to 512, with a final fully connected classification layer serving as the last layer. As compared to the traditional method (95.6% accuracy) and with fewer false positives, the LWR model performed very well (97.26–98.11%). With the goal of improving PlantVillage's leaf disease classification system, Hukkeri et al. [9] developed the EfficientNet architecture. EfficientNet got the best accuracy of 97.5% by using pre-trained CNNs such InceptionV3, ResNet50, MobileNetV2, AlexNet, and ViperNet. It did better than other models even when the input photos were enlarged to 132×132. By evaluating different CNN variations such as GCNN, ShuffleNet, and depth-wise separable CNN, Rizwan et al. [10] created a CNN-based method for autonomous plant disease diagnosis. The suggested ShuffleNet model found a happy medium between accuracy and efficiency, with an average success rate of 96.66%, in contrast to MobileNetV2 and Xception, which were more accurate but required more processing resources. Using a pre-trained CNN with gradient-based iterative correction and data augmentation to decrease overfitting, Wang et al. [11] presented a deep TL method for plant leaf disease identification. With less training time, their technique obtained an accuracy of over 95% on the PlantVillage dataset. Dey et al. [12] utilized CNN with TL to identify plant diseases, employing the PlantVillage dataset, with 80% allocated for training and 20% for testing. The revised AlexNet attained superior performance, with 96.63% accuracy, 92% precision, 91% recall, and 91% F1-score, surpassing both VGG16 and VGG19. Rastogi et al. [13] employed CNNs with categorical cross-entropy loss and the Adam optimizer to detect plant diseases in several classes early on. Their model identified 12 illnesses in tomato, potato, and bell pepper leaves with 86.21% accuracy after being trained for 20 epochs using data augmentation on the PlantVillage dataset.

Table I also includes a comparison of the latest studies published on the PlantVillage dataset that use the TL technique for disease classification in plants. These studies were published in various journals. Our DL-based ensemble method outperformed all previous studies when measured against a set of 15 data classes using the TL approach. The suggested ensemble model has the potential to play a pivotal role in the early diagnosis of crop diseases and makes a substantial contribution to our study.

III. Research Methodology

This section outlines our proposed method for accurately detecting and classifying the plant diseases. The suggested DL-based ensemble model is tested against well-known CNN models using a XAI technique to determine how well it performs in crop image categorization. Figure 1 (A) featured our training process, Figure 1 (B) showcased our testing phase, and Figure 1 (C) showcased our TL technique to evaluating the best model.

A. Data Augmentation

Image data generators are crucial for training DL models since they increase the dataset's diversity and dynamic augmentation, which fortify the model. For our study, we used the image data generator for the Keras model. We employed 45-degree left and right rotation, 30% zoom, and flips in the horizontal and vertical directions to apply augmentation to photographs. The example of image improvement is shown in Figure 2.

B. Image Data Preprocessing

The DL models' performance is enhanced during the preprocessing stage. The model's usefulness is enhanced since it helps with standardizing and normalizing pixel values, reducing noise, and improving features. We utilized a cropping and resizing method that reduced the image to 224 × 224 pixels while preserving any noticeable disease spots in order to guarantee homogeneity.

C. Dataset Splitting

After data preparation, we partitioned the dataset into training, validation, and test sets. Here are the percentages: 80% is used for training, 10% for validation, and 10% for testing the whole dataset. A total of 20,639 photos from 15 classes in the well-known PlantVillage [14] dataset were used in this experiment. Using the Plantvillage dataset, we more accurately detect a variety of crop leaf diseases, including pepper bacterial spot (Pbs), pepper bell healthy (Pbh), potato early blight (Peb), potato healthy (Ph), potato late blight (Plb), tomato target spot (Tts), tomato mosaic virus (Tmv), tomato curl virus (Tcv), tomato bacterial spot (Tbs), tomato early

TABLE I: Comparison of the proposed method with existing studies that employed transfer learning (TL) and data augmentation techniques for crop disease detection on the PlantVillage dataset.

Previous Study	Dataset	TL Used	No. of Classes	Best Method	Accuracy(%)
Khalid *et al.* [5]	PlantVillage	No	20	MobileNet	96.00
Zhang *et al.* [6]	PlantVillage	Yes	6	SE-SK-CapResNet18	98.58
Radočaj *et al.* [7]	PlantVillage	Yes	6	InceptionV3 with IncMB	97.78
Sadhasivam *et al.* [8]	PlantVillage	No	4	Lightweight Residual (LWR)	97.26–98.11
Hukkeri *et al.* [9]	PlantVillage	No	38	EfficientNet	97.50
Rizwan *et al.* [10]	PlantVillage	Yes	38	ShuffleNet	96.86
Wang *et al.* [11]	PlantVillage	Yes	38	VGG16	95.00
Dey *et al.* [12]	PlantVillage	No	38	AlexNet	96.63
Rastogi *et al.* [13]	PlantVillage	Yes	15	CNN	86.21
This Work	**PlantVillage**	**Yes**	**15**	**Proposed ensemble model**	**98.70**

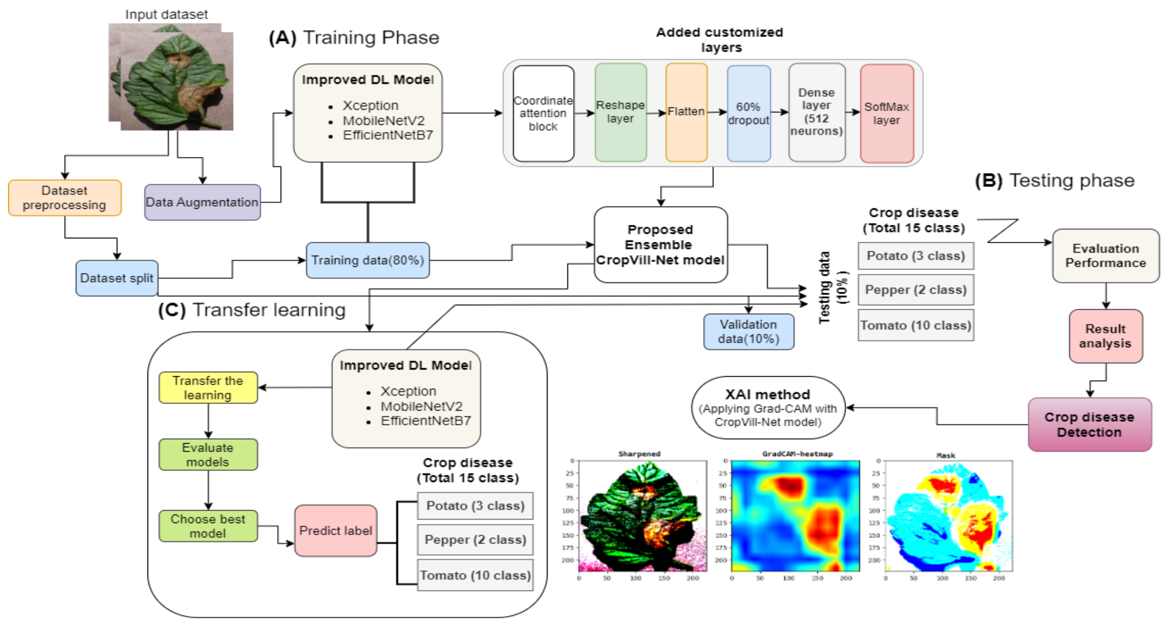

Fig. 1: Workflow methodology.

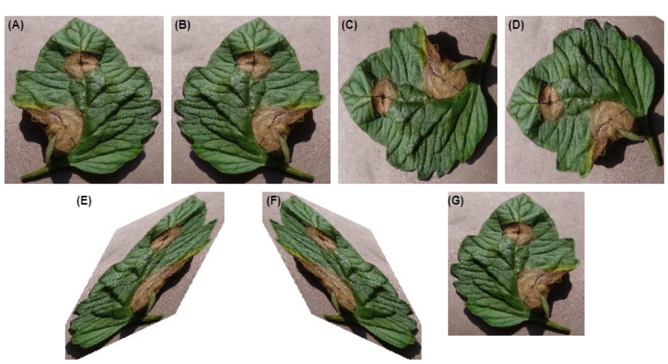

Fig. 2: Examples of data augmentation applied to tomato leaf images: (A) horizontal flip (left), (B) horizontal flip (right), (C) vertical flip (up), (D) vertical flip (down), (E) rotation by 45° (right), (F) rotation by 45° (left), and (G) zoom by 30%.

blight (Teb), tomato healthy (Th), tomato late blight (Tlb), tomato leaf mold (Tlm), tomato septoria spot (Tss), and tomato spider mite (Tsm).

D. DL models

1) Xception: Depthwise separable convolution layers are used in Xception, a deep convolutional architecture. In this sense, an inception module with the maximum number of towers is equivalent to a depthwise separable convolution. Based on this finding, we suggested a unique deep CNN architecture [15], in which depthwise separable convolutions are used as part of the inception modules. Data inputted with a single *1* × *1* convolution size yields various *3* × *3* convolution sizes in the absence of average pooling. Different parts of the output channels are used by these convolution sizes before forward patterns are obtained.

2) MobileNetV3: Small restraint layers appear for the input and output of the leftover section in the MobileNetV3 model. In addition, MobileNetV3 refines attributes in the headway by the help of light convolutions. It helps to take off non-linearities in the narrowing surfaces [16]. With O_t and I_t defines input and output passage deepness, sequentially, MobileNetV3's depthwise convolutions regulates input dimensions ($I_d \times I_d \times O_t$) and output dimensions ($D_G \times D_G \times I_t$) separately of average convolutional surfaces [17]. MobileNetV3 is prominent for its great mechanism, which utilizes independent layers that are responsible for feature pulling out and gathering, which helps to improve its computing efficiency. Inside the downward convolution layer, the depthwise convolution particular kernel is represented as $\hat{C}$, and its dimensions are as follows: ($I_c \times I_c \times O_t$).

3) EfficientNetB7: The EfficientNetB7 [18] model performs better on problems involving CV. Our improved classification model and increased accuracy are the result of utilizing the EfficientNetB7 model's changed structure. After more optimization with additional layers, we settled on the EfficienNetB7 model. With *3 x 3* convolutions, reduced specification, and full utilization of modern accelerators, the deepness section is cutting edge. Convolution *1 × 1* for enlargement in MB Convolution and Convolution *3 × 3* for deepenings make up the starting portion. Afterwards, a layer of average pooling was added to the structure to increase its strength. Effective and precise neural network topologies are the product of a mathematical process that involves stabilizing all of the parameters. The foundation of EfficientNetB7's operation is a composite metric that controls the width, height, and number of channels per surface in the convolutional parts, as well as the number of surfaces and channels, and the image resolution. To improve the accuracy of the related architecture, we use four channels in the deepening convolution.

4) Proposed Ensemble Method: We suggest an ensemble [19] model that uses the aforementioned DL models to achieve finer results. In our ensemble design, Xception, MobileNetV3, and EfficientNetB7 all contribute to the final estimation. When we combine these three models, we get a DL ensemble that does quite well when it comes to crop disease classification. The input data for the state-of-the-art ensemble model is available first. Following this, the preprocessed pictures were merged using the three DL architectures mentioned before. Up top, you can see that the model accounts covered every single layer and convolution region. Since then, we have improved our capacity to identify crop diseases by adding a few layers to our design, keeping it up to date. Initially, a correlate section was attached. Then, a reshaped and flattened layer was put in. Tensors that have multifaceted input are flattened via the flattened layer. In the lead, the data are sent to the following layer, where they are altered into a 1-D array. Next, a 60% dropout layer was attached. At last, a softmax layer is included to drive the neurons. With each training cycle, this dropout layer prunes the model by removing half of the neuron links. Next, a thick layer of 512 neurons is inserted. In this case, we used the RELU activation function. The fifteen classes in the Plantvillage dataset are being classified by fifteen neurons in the last layer. Multiclass classification is carried out here using the Softmax activation function. Lastly, in order to forecast the true category, a global average pooling layer is incorporated.

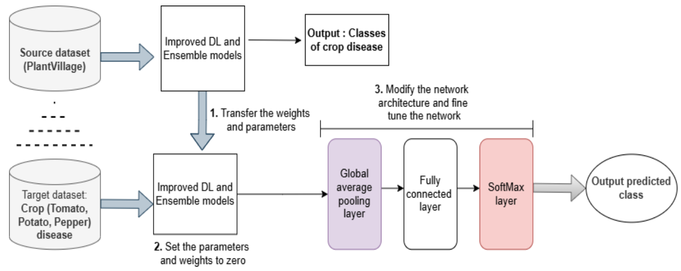

Fig. 3: Process of the transfer learning (TL) method. Pretrained weights learned from large-scale datasets are transferred to the target task, followed by fine-tuning using global average pooling, fully connected, and Softmax layers.

E. Hyperparameter tuning for the proposed ensemble model

We used hyperparameter tuning to fine-tune our proposed ensemble model, and then we put it to use to identify crop diseases. Model performance and generalizability can be optimized with its help. Optimizing models involves modifying hyperparameters like learning rate, number of hidden layers, and dropout rate in order to enhance accuracy and decrease problems like underfitting or overfitting. We utilized the best learning rate of $2e-3$ and kernel of [3 x 3] for the agricultural disease detection task. The 64 batch size and the 0.60 dropout rate are also included. We used the Adam optimizer to get the accurate answer for the multiclass classification problem.

F. Transfer learning using DL models

The TL technique [20], which uses a model learned for a specific task as a baseline for a different one, has been employed in this research to improve the accuracy of DL models. This can be in a different training procedure with less data. As the source model for TL, a CNN model that has been trained on a large amount of data and is capable of handling comparable types of data should be ready before beginning the TL process.

G. Gradient-weighted Class Activation Mapping (Grad-CAM)

Grad-CAM [21] generates a heatmap that highlights key areas for prediction using the gradients of the outputs from the convolution layers of the proposed ensemble model. This approach improves the model's interpretability by providing visual explanations for the network's decisions.

IV. Results and Discussion

A. Specifications of the environment setup

The computational power required for picture analysis and classification can be supplied in large part by graphics processing units (GPUs). An additional piece of hardware and a higher price tag are necessary for a GPU installation to allow processing activities. For this reason, we trained our model on the Google Colab platform, which provides access to powerful Cloud GPUs. In Table II, we have detailed the settings and parameters of the research environment.

TABLE II: Parameters and specifications requirements.

Environment	Parameters
Platform	Google Colab
Language version	Python 3.0
Graphics card	TPU, 13.7 GB
Available RAM	14.3 GB
Disk space	21.9 GB

B. Results Analysis

Every model that was used in this investigation and its findings are given below. The results indicated that our proposed model was employed to assess the models' robustness. Furthermore, this study evaluated and compared the efficacy of our recommendations using Xception, EfficientNetB7, MoileNetV2, and InceptionV3. Regarding the crop disease identification, Table V represents the experimental results of our applied models, the suggested ensemble model outperforms the other models significantly. As DL models can only use pretrained data to TL, they may not be able to adapt to new data as well and may even be prejudiced. Group weights and the TL approach make our proposed model more responsive to changes in the data or model architecture, which makes overfitting easily addressed. When compared to DL models, the ensemble model frequently has a better accuracy rate (98.70%).

TABLE III: Classification results using the TL approach and augmentation technique on the test set.

Method	**Precision (%)**	**Recall (%)**	**F1-score (%)**
Proposed model	99.10	98.70	98.80
Xception	96.70	96.40	96.50
MobileNetV2	95.30	93.50	93.50
EfficientNetB7	93.10	91.10	91.20
InceptionV3	89.60	85.40	85.80

The comprehensive report that utilized the TL strategy and augmentation technique during the testing phase of crop disease classification is presented in Table III. This report takes into account the precision, recall, and F-1 score for each approach. In this investigation, our proposed model outperformed all other models. With a recall of 96.70%, an F1 score of 98%, and a precision of 99.10%, this strategy proved to be incredibly successful. Furthermore, with a precision of 96.70%, Xception achieves the second-highest level.

TABLE IV: Classification results without TL approach and augmentation technique on the test set.

Method	**Precision (%)**	**Recall (%)**	**F1-score (%)**
Proposed model	98.00	97.50	97.80
Xception	96.00	96.20	96.00
MobileNetV2	94.80	93.00	93.20
EfficientNetB7	91.70	91.80	90.40
InceptionV3	86.60	84.00	84.50

Table IV displays the detailed report of crop disease classification for each method without the TL approach and augmentation technique. The precision of our proposed ensemble model decreased by 1% in this case. Despite the absence of the TL approach and augmentation procedures, the ensemble method produced an F1 score of 97.80%, a precision of 98.00%, and a recall of 97.50%.

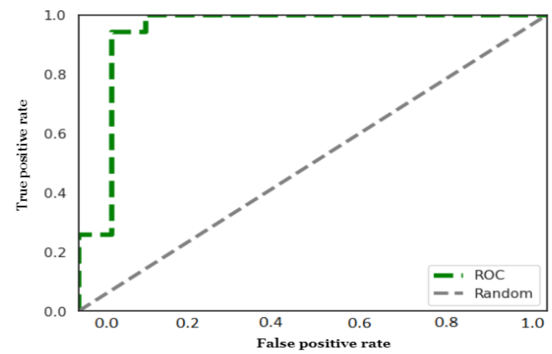

Fig. 4: ROC curves for the proposed model using the TL approach with data augmentation.

Using the TL approach and augmentation techniques in the testing phase, the ROC curve is displayed in Figure 4, which is an additional compliance indicator for the ensemble model. It has been noted that the ensemble model that has been proposed possesses an exceptional micro average AUC score of more than 98%, which is the greatest number that has ever been officially recorded. The proposed method shows outstanding success in recognizing symptoms of crop disease within the scope of the technology that was proposed previously.

C. Output analysis of Grad-CAM and Comparative study

Grad-CAM analysis is a technique that can be utilized to examine the appearance of regions within an image that are the primary focus of a DL model. Our suggested ensemble model, which incorporates three improved DL models, was employed in this study to build maps of afflicted plants. Figure 5 shows that the Grad-CAM in the proposed ensemble model allows for a highly smooth detection of the impacted region using crop disease images.

V. CONCLUSION AND FUTURE WORK

Plant disease classification at an early stage is critical for enhancing crop yields and meeting food demands. Although much research has been conducted on plant leaf disease classification, our proposed ensemble model performed better than the approaches reported in the literature. We evaluated the performance of the model with and without TL and data augmentation strategies. Using TL and data augmentation techniques, our proposed model achieves a maximum accuracy of 98.70%, whereas, without TL and data augmentation, it achieves 97.50%. The experimental findings demonstrated that the proposed ensemble model is exceptionally reliable in its ability to accurately detect nearly all diseases affecting plant leaves, as evidenced by its high accuracy.

TABLE V: Loss and Accuracy for all evaluated models using the validation set in this research.

Type	Approach	Proposed model	Xception	MobileNetV2	EfficientNetB7	InceptionV3
Accuracy (%)	Using TL approach and augmentation	**98.70 (Proposed)**	96.40	93.50	91.10	85.40
	Without TL approach and augmentation	97.50	96.20	93.00	90.50	84.20
Loss (%)	Using TL approach and augmentation	0.80	1.79	2.67	3.32	4.15
	Without TL approach and augmentation	1.38	2.10	2.88	3.77	4.27

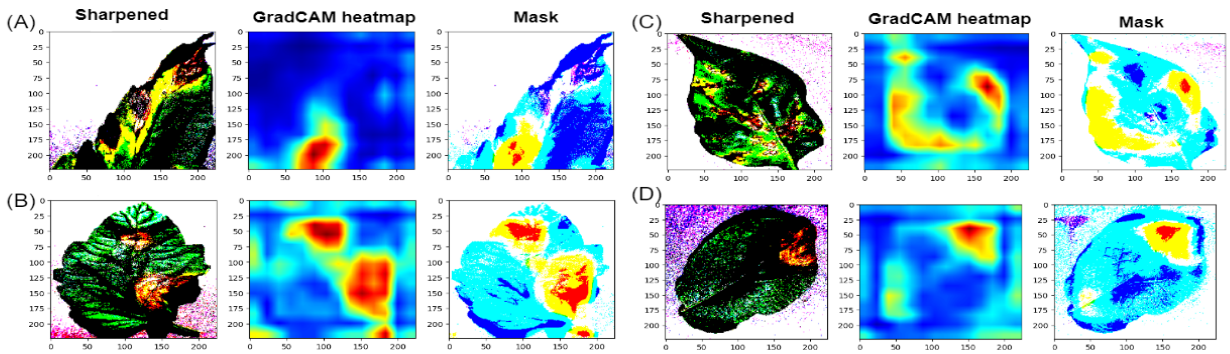

Fig. 5: Representative examples from the PlantVillage dataset showing the input images, sharpened images, Grad-CAM heatmaps, and corresponding mask visualizations generated by the proposed ensemble model for (A) Teb, (B) Tlb, (C) Pbs, and (D) Plb classes.

There were limitations on the chosen classes from the PlantVillage collection because the majority of the photos were authentic, while some had been altered. A number of real-world variables, including climate change, will affect the final product. The impact of climatic change on plant diseases should be studied in future studies, and a variety of photos of damaged leaves should be collected.

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